

COS30018 – OPTION ‘C’

TASK ‘3’ REPORT

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1. Environment Setup:

I installed a Python virtual environment with venv. Numpy, pandas, matplotlib, scikit-learn, yfinance, mplfinance were installed as core libraries. The yahoo_fin package was first tried and provided repeated errors (JSONDecodeError, 429 responses). Thus, I substituted it with yfinance as it was more stable to download stock data. The data sets of Apple Inc. (AAPL) were collected between January 2023 and July 2025 on a daily basis.

2. Implementation:

The project workflow was visually oriented and not prediction model. Key components included:

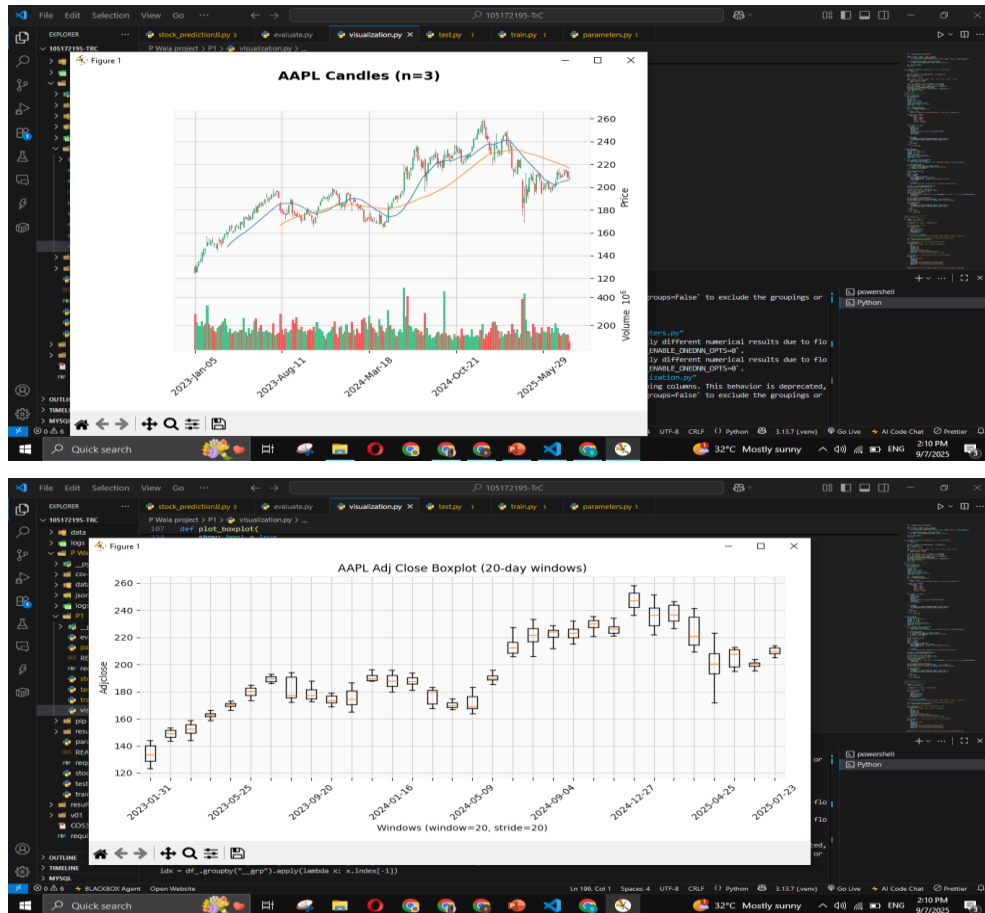
- Data Loader: Strong data search based on yfinance.download().
- Chart: candlestick chart of short- and long-term moving averages.
- Boxplot Analysis: Adjusted Close values in 20-day rolling windows.
- Modular Functions: To download data, preprocess, feature engineer and plot.

3. Testing and Results:

The tests were made on AAPL historical data. Results included:

- Candlestick Chart: displayed the overall increase till mid 2024 (peak of around \$260) and then a correction and stabilization around 200-220.
- Boxplot: Median Adj Close is increasing, as in the early of 2023 it was around \$140, and at the end of 2024, around 230-240. Expansive distributions in mid-2024 were volatility.
- Outliers: Market prices which have been captured by both dips and spikes.

Generally, the outputs had a steady correspondence with actual historical market action.



4. Comparison with Preliminary Attempts:

The first efforts with `yahoo_fin` were often unsuccessful because of rate limiting and unreliable API responses. The replacement with `yfinance` made the results more reliable and reproducible. This version offered professional grade visualizations, as compared to the earlier versions of the exploratory output.

5. Conclusion:

Visualization process was useful in the analysis of AAPL price movements. Trend and volatility indicators were both brought out by a combination of candlestick plots and boxplots. The pipeline is also extensively modular, which suits its use with predictive models (e.g., LSTM, GRU). This provides a platform on the higher levels of financial forecasting activities.

Appendix:

- Checking of environment (pip list, requirements.txt)
- console output of yfinance that data is retrieved is valid.
- Moving averages candlestick chart.
- Boxplot of Adjusted Close prices.
- CSV files of processed data that were exported.